

Rotation Matrices — Everything in One Place

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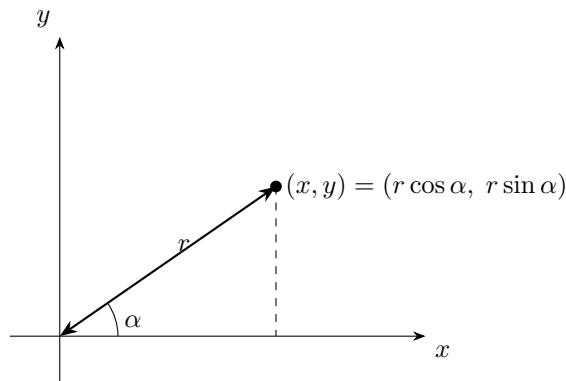
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1 The 2D rotation (the engine)

1.1 Goal and polar setup

We want a matrix $R(\theta)$ so that multiplying (x, y) gives the point rotated counterclockwise by θ about the origin. Write (x, y) in polar form $x = r \cos \alpha$, $y = r \sin \alpha$ (radius r , angle α). Rotating by θ puts the point at angle $\alpha + \theta$ with the same r :

$$x' = r \cos(\alpha + \theta), \quad y' = r \sin(\alpha + \theta).$$



1.2 Apply the angle-addition identities

Using $\cos(\alpha + \theta) = \cos \alpha \cos \theta - \sin \alpha \sin \theta$ and $\sin(\alpha + \theta) = \sin \alpha \cos \theta + \cos \alpha \sin \theta$,

$$x' = r \cos \alpha \cos \theta - r \sin \alpha \sin \theta = x \cos \theta - y \sin \theta,$$

$$y' = r \sin \alpha \cos \theta + r \cos \alpha \sin \theta = x \sin \theta + y \cos \theta,$$

using $x = r \cos \alpha$, $y = r \sin \alpha$. Collecting into a matrix:

$$R_{2D}(\theta) = \begin{pmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{pmatrix}$$

1.3 Two worth-noting facts

- The columns are unit vectors (orthonormal): $R^T R = I$, $\det R = 1$.
- Rotation preserves length and angle — why the L^2 norm and dot product are the “rotation-friendly” tools. Same θ as in $\|u\|\|v\| \cos \theta$.

2 2D rotation *is* rotation about the z -axis

A rotation in the xy -plane is exactly a rotation in $\mathbb{R}^3$ about the z -axis, with every point’s z fixed. Embed the 2D matrix by appending the z -axis entries so z is untouched:

$$R_z(\theta) = \begin{pmatrix} \cos \theta & -\sin \theta & 0 \\ \sin \theta & \cos \theta & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

Multiplying (x, y, z) : the x, y part spins by the 2D block, and $z' = z$. The z -axis is real, not imaginary — it is the third direction perpendicular to the 2D picture (“out of the paper”).

3 The same recipe: R_y and R_x

By identical reasoning, the pivot axis gets a $(0 \dots 1 \dots 0)$ row/column and the other two coordinates get the 2D block:

$$R_y(\phi) = \begin{pmatrix} \cos \phi & 0 & \sin \phi \\ 0 & 1 & 0 \\ -\sin \phi & 0 & \cos \phi \end{pmatrix}, \quad R_x(\psi) = \begin{pmatrix} 1 & 0 & 0 \\ 0 & \cos \psi & -\sin \psi \\ 0 & \sin \psi & \cos \psi \end{pmatrix}.$$

4 The easy way to derive R_x and R_y

You reuse the 2D result in each perpendicular plane, just with different letters playing “ x and y .”

4.1 R_x = rotate the yz -plane (pivot x , untouched)

The plane’s axes $\{y, z\}$ already read in coordinate order, so the 2D formula applies directly:

$$y' = y \cos \psi - z \sin \psi, \quad z' = y \sin \psi + z \cos \psi, \quad x' = x,$$

$$R_x(\psi) = \begin{pmatrix} 1 & 0 & 0 \\ 0 & \cos \psi & -\sin \psi \\ 0 & \sin \psi & \cos \psi \end{pmatrix}.$$

4.2 R_y = rotate the zx -plane (pivot y , untouched)

Here the plane’s natural order is $\{z, x\}$; apply 2D to z (first) and x (second):

$$z' = z \cos \phi - x \sin \phi, \quad x' = z \sin \phi + x \cos \phi, \quad y' = y,$$

then rewrite in (x, y, z) order:

$$x' = x \cos \phi + z \sin \phi, \quad y' = y, \quad z' = -x \sin \phi + z \cos \phi,$$

$$R_y(\phi) = \begin{pmatrix} \cos \phi & 0 & \sin \phi \\ 0 & 1 & 0 \\ -\sin \phi & 0 & \cos \phi \end{pmatrix}.$$

4.3 Why the sign flip in R_y is not arbitrary — the ordering story

The key rule: the 2D block $\begin{pmatrix} \cos & -\sin \\ \sin & \cos \end{pmatrix}$ applies correctly only to the pair that makes a *right-handed* frame with the pivot axis (i.e. the pair whose cross product points along positive pivot axis):

Rotation about	Plane axes	right-handed pair?
R_z	x, y	$x \times y = +z \checkmark$
R_x	y, z	$y \times z = +x \checkmark$
R_y	x, z	$x \times z = -y \text{ ✗, but } z \times x = +y \checkmark$

The correct ordered pair for R_y is (z, x) , not (x, z) — read “ z is followed by x ” *forward* (cyclic order $x \rightarrow y \rightarrow z \rightarrow x$). Applying the standard block to (z, x) :

$$\begin{pmatrix} z' \\ x' \end{pmatrix} = \begin{pmatrix} \cos & -\sin \\ \sin & \cos \end{pmatrix} \begin{pmatrix} z \\ x \end{pmatrix},$$

which puts $-\sin$ in the (row z , col x) slot. Displaying that in the full (x, y, z) matrix: row z is the **bottom** row and col x is the **first** column, so $-\sin$ lands **bottom-left**.

Why it *looks* inverted (the “order swapped” resolution). If you wrote the block in (x, z) order (treating x as “first”), $x \times z = -y$ gives a *left-handed* pair and you’d wrongly get $-\sin$ top-right. The apparent “inversion” is purely: the plane’s natural right-handed pair (z, x) does not match the display order (x, y, z) , so the two off-diagonal entries swap places. Cropping R_y to its xz sub-block gives $\begin{pmatrix} \cos & \sin \\ -\sin & \cos \end{pmatrix}$ — the same block, just with x and z roles, hence rows/columns, exchanged vs. how R_z ’s block looks.

Cyclic relabeling: R_y is R_z in disguise. Cyclically rename $z \rightarrow x, x \rightarrow y, y \rightarrow z$ and “rotate the zx -plane about y ” becomes “rotate the xy -plane about z ”:

$$R_y(\phi) = \begin{pmatrix} \cos & 0 & +\sin \\ 0 & 1 & 0 \\ -\sin & 0 & \cos \end{pmatrix}_{(x,y,z)} \xrightarrow{(z,x,y)} \begin{pmatrix} \cos & -\sin & 0 \\ \sin & \cos & 0 \\ 0 & 0 & 1 \end{pmatrix}_{(z,x,y)},$$

which is R_z -shaped. So nothing is special about R_y — it only *seems* inverted because the display order (x, y, z) is out of step with the rotation’s natural pair (z, x) .

Never-perform reliability check at $\phi = 90^\circ$. A 90° rotation about y sends $z \rightarrow +x$ and $x \rightarrow -z$. So column 3 (image of z) must be $(1, 0, 0)$, and column 1 (image of x) must be $(0, 0, -1)$: $R_y(90^\circ) = \begin{pmatrix} 0 & 0 & 1 \\ 0 & 1 & 0 \\ -1 & 0 & 0 \end{pmatrix}$, which reproduces the matrix exactly. This physical axis check recovers R_y without pattern-matching a sign table.

5 Roll, pitch and yaw (the human frame)

Body frame: x = forward (nose), y = out the right, z = down.

Name	Axis	Matrix	Rotates the plane
Yaw	z	$R_z(\theta)$	xy
Pitch	y	$R_y(\phi)$	zx
Roll	x	$R_x(\psi)$	yz

6 Order matters: Euler angles and the ZYX convention

6.1 There is no single universal order

The order is a convention, and 12 intrinsic orders exist. Two families:

- **Proper Euler angles:** same axis for 1st and 3rd rotation (e.g. ZXZ, ZYZ) — classical mechanics / quantum spin.
- **Tait–Bryan angles:** three different axes (e.g. ZYX, XYZ) — aerospace / navigation / robotics. Roll–pitch–yaw is Tait–Bryan.

Euler angles are also *non-unique* and subject to **gimbal lock** (a lost degree of freedom when the middle angle aligns two axes) — why robotics often uses quaternions.

6.2 ZYX (intrinsic) order = yaw, then pitch, then roll

Each subsequent rotation happens about the axis that has just been rotated (body frame):

1. **Yaw:** about the body’s z -axis, by θ .
2. **Pitch:** about the body’s *new* y -axis, by ϕ .
3. **Roll:** about the body’s *new* x -axis, by ψ .

The total rotation is the right-to-left product

$$R = R_x(\psi) R_y(\phi) R_z(\theta)$$

applied as $v' = Rv$.

6.3 Why the matrix closest to the vector acts first

Matrix–vector multiplication groups right-to-left: $Rv = R_x(R_y(R_z v))$, so $R_z v$ is computed first, then R_y , then R_x — matching intrinsic ZYX. Since v sits on the right of $R_x R_y R_z$, the **first-applied matrix goes on the right**:

“ZYX” describes *application* order (Z first); the matrix string mirrors it by putting the first-applied factor rightmost. Write left-to-right, read right-to-left. Swapping to $R_z R_y R_x$ yields the reversed (extrinsic) reading.

6.4 Fixed vs rotating axes

Rotating about the *moving* axes (intrinsic ZYX) gives the *same* final geometry as rotating about the *fixed* axes in reversed order:

$$\text{intrinsic } z, y, x \quad \Longleftrightarrow \quad \text{extrinsic } x, y, z.$$

7 Complete table and summary

Rotation	Matrix	Plane
$R_z(\theta)$	$\begin{pmatrix} \cos \theta & -\sin \theta & 0 \\ \sin \theta & \cos \theta & 0 \\ 0 & 0 & 1 \end{pmatrix}$	xy
$R_y(\phi)$	$\begin{pmatrix} \cos \phi & 0 & \sin \phi \\ 0 & 1 & 0 \\ -\sin \phi & 0 & \cos \phi \end{pmatrix}$	zx
$R_x(\psi)$	$\begin{pmatrix} 1 & 0 & 0 \\ 0 & \cos \psi & -\sin \psi \\ 0 & \sin \psi & \cos \psi \end{pmatrix}$	yz

- 3D rotation about any axis = the 2D rotation in the perpendicular plane, with that axis left untouched.
- R_z is literally the 2D matrix plus a 0, 0, 1 row/column.
- The sign in each matrix lives at “first-row/second-column of the block” — it follows the plane’s axes, never an arbitrary choice.
- General 3D orientation = an *ordered* product; ZYX means z (yaw), then y (pitch), then x (roll), about the already-rotated axes.